YeshwantraoChavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
Accredited 'A' Grade by NAAC
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Department of Applied Mathematics & Humanities

Electronic Question Answer Bank

Course Name: Calculus and Vector **Course Code: 23GE1101**

UNIT-VI : Vector Integration & Applications

Sr.No.	Unit No.	Text of Question	Time required in Minutes	Marks allotted	lengthiness	Difficulty level	CO Mapping	Type Descriptive / Numerical
1	Unit 6	If $\bar{r}(t) = 5t^2\hat{i} + t\hat{j} - t^3\hat{k}$ Prove that $\int_1^2 \left(\bar{r} \times \frac{\overline{d^2r}}{dt^2}\right) dt = -14\hat{i} + 75\hat{j} - 15\hat{k}$.	10	5	Low	Easy	CO-4	Numerical

		$\bar{r} = 5t^2 i + t j - t^3 k$ $\frac{d\bar{r}}{dt} = 10t i + j - 3t^2 k$ $\frac{d^2\bar{r}}{dt^2} = 10 i + 0 j - 6t k$ Now, $\bar{r} \times \frac{d^2\bar{r}}{dt^2} = \begin{vmatrix} i & j & k \\ 5t^2 & t & -t^3 \\ 10 & 0 & -6t \end{vmatrix}$ $= i[-6t^2] - j[-30t^3 + 10t^3] + k[0 - 10t]$ $= -6t^2 i + (20t^3) j - 10t k.$ $\therefore \int_1^2 \left[\bar{r} \times \frac{d^2\bar{r}}{dt^2}\right] dt = \int_1^2 [-6t^2 i + 20t^3 j - 10t k]\, dt$ $= -6\left[\frac{t^3}{3}\right]_1^2 i + 20\left[\frac{t^4}{4}\right]_1^2 j - 10\left[\frac{t^2}{2}\right]_1^2 k$ $= -2(8-1) i + 5(16-1) j - 5(4-1) k$ $\therefore \int_1^2 \left[\bar{r} \times \frac{d^2\bar{r}}{dt^2}\right] dt = -14 i + 75 j - 15 k.$						
2	Unit 6	Evaluate $\int \bar{F}.\overline{dr}$ from the point A(0,0,0) to B(1,1,1) at along the curve $\bar{r} = i\,t + j\,t^2 + k\,t^3$ given $\bar{F} = xy\,i - z^2\,j + xyz\,k.$	10	5	Low	Easy	CO-4	Numerical

Soln. We have,

$$\bar{r} = xi + yj + zk = ti + t^2 j + t^3 k$$

$$\therefore \frac{d\bar{r}}{dt} = i + 2tj + 3t^2 k$$

Since $\bar{r}$ is the position vector of points on the curve, the coordinates of any point in terms of t are given by

$$x = t,\ y = t^2,\ z = t^3$$

From these equations, we find that the points A (0, 0, 0) and B (1, 1, 1) correspond to $t = 0$ and $t = 1$.

Also,

$$\bar{F} = xy\,i - z^2 j + xyz\,k$$

$$= t^3 i - t^6 j + t^6 k$$

$$\therefore \int_A^B \bar{F} \cdot d\bar{r} = \int_{t=0}^{t=1} \bar{F} \cdot \frac{d\bar{r}}{dt}\,dt$$

$$= \int_0^1 (t^3 i - t^6 j + t^6 k) \cdot (i + 2tj + 3t^2 k)\,dt$$

$$= \int_0^1 (t^3 - 2t^7 + 3t^8)\,dt$$

$$= \left[\frac{t^4}{4} - \frac{t^8}{4} + \frac{t^9}{3}\right]_0^1$$

$$= \frac{1}{4} - \frac{1}{4} + \frac{1}{3}$$

$$\therefore \boxed{\int_A^B \bar{F} \cdot d\bar{r} = \frac{1}{3}} \quad \textbf{Ans.}$$

3	Unit 6	Find the work done in moving a particle once round the circle $x^2 + y^2 = 9, z = 0$ under the field of force given by $\bar{F} = (2x - y + z)i + (x + y - z^2)j + (3x - 2y + yz)k$. Ans. We have, $\bar{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + yz)\hat{k}$ and Work done $= \int_C \bar{F} \cdot d\bar{r}$ $= \int_C [(2x - y + z)i + (x + y - z^2)j + (3x - 2y + yz)\hat{k}] \cdot (dx\, i + dy\, j + dz\, k)$ $= \int_C [(2x - y)\, dx + (x + y)\, dy]$ $[\because z = 0, \Rightarrow dz = 0]$ The parametric equations of given path $x^2 + y^2 = 9$ are $x = 3\cos\theta$, $y = 3\sin\theta$ where θ varies from 0 to 2π $\therefore dx = -3\sin\theta d\theta$ $dy = 3\cos\theta\, d\theta$ $\therefore$ Work done $= \int_0^{2\pi} [2(3\cos\theta) - 3\sin\theta](-3\sin\theta\, d\theta) + (3\cos\theta + 3\sin\theta)(3\cos\theta\, d\theta)$ $= \int_0^{2\pi} [(6\cos\theta - 3\sin\theta)(-3\sin\theta\, d\theta) + (3\cos\theta + 3\sin\theta)(3\cos\theta\, d\theta)]$ $= \int_0^{2\pi} [-18\sin\theta\cos\theta + 9\sin^2\theta + 9\cos^2\theta + 9\cos\theta\sin\theta]\, d\theta$ $= \int_0^{2\pi} [9(\sin^2\theta + \cos^2\theta) - 9\sin\theta\cos\theta]\, d\theta$ $= 9\int_0^{2\pi} (1 - \sin\theta\cos\theta)\, d\theta$ $= 9\left[\theta - \frac{\sin^2\theta}{2}\right]_0^{2\pi}$ $= 9[2\pi - 0]$ $\therefore$ **Work done $= 18\pi$** Ans.	10	5	Low	Easy	CO-4	Numerical

4	Unit 6	A vector field is given by $\bar{F}$ = siny $\hat{\imath}$ + x (1+ cosy) $\hat{\jmath}$ Evaluate the line integral over the circular path given by $x^2 + y^2 = a^2$; z=0. Soln. We have, $\bar{F} = \sin y\ i + x\ (1+\cos y)\ j$ and Work done $= \int_C \bar{F}.d\bar{r}$	10	5	Low	Easy	CO-4	Numerical

$$= \int_C [(\sin y)i + x(1+\cos y)j].[dxi + dyj]$$

$$[\because z = 0, \Rightarrow dz = 0]$$

$$\therefore \int_C \bar{F}.d\bar{r} = \int_C [\sin y\,dx + x(1+\cos y)\,dy]$$

$$= \int_C [d(x \sin y) + x\,dy]$$

where d is differential operator.

The parametric equations of given path

$x^2 + y^2 = a^2$ are $x = a\cos\theta$, $y = a\sin\theta$,

where θ varies from 0 to 2π.

$$\therefore \int_C \bar{F}.d\bar{r} = \int_0^{2\pi} \{d[a\cos\theta \sin(a\sin\theta)] + a\cos\theta . a\cos\theta\, d\theta\}$$

$$= \int_0^{2\pi} \{d[a\cos\theta \sin(a\sin\theta)] + a^2\cos^2\theta . d\theta\}$$

$$= [a\cos\theta \sin(a\sin\theta)]_0^{2\pi} + a^2 \int_0^{2\pi} \cos^2\theta\, d\theta$$

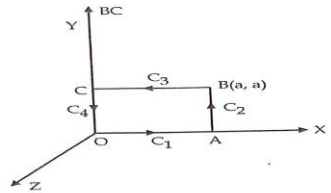
$$= 0 + a^2 \int_0^{2\pi} \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$$= \frac{a^2}{2}\left[\theta + \frac{\sin 2\theta}{2}\right]_0^{2\pi}$$

$$= \frac{a^2}{2}.2\pi .$$

$$= \pi a^2$$

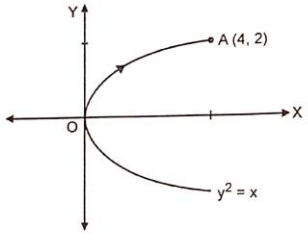
$\therefore$ $\boxed{\text{Work done} = \pi a^2}$ **Ans.**

5	Unit 6	Evaluate $\int_c \bar{F}.\, d\bar{r}$, Where c is the boundary of the square in the plane z = 0 and bounded by then lines x = 0, y = 0, x = a and y = b. **Soln. Given :** $\bar{F} = x^2 i + xy\, j$ $\int_C \bar{F}.\overline{dr} = \int_{OA} \bar{F}.\overline{dr} + \int_{AB} \bar{F}.\overline{dr} + \int_{BC} \bar{F}.\overline{dr} + \int_{CO} \bar{F}.\overline{dr}$ ..... (1) Here, $\bar{r} = xi + yj \Rightarrow d\bar{r} = dxi + dyj$ $\therefore \bar{F}.d\bar{r} = x^2 dx + xydy$  On OA, y = 0 $\therefore \bar{F}.d\bar{r} = x^2 dx$ $\Rightarrow \int_{OA} \bar{F}.d\bar{r} = \int_0^a x^2 dx = \left[\frac{x^3}{3}\right]_0^a = \frac{a^3}{3}$ On AB, x = a $\therefore$ dx = 0 $\therefore \bar{F}.d\bar{r} = ay\, dy$ $\Rightarrow \int_{AB} \bar{F}.d\bar{r} = \int_0^a ay\, dy = a\left[\frac{y^2}{2}\right]_0^a = \frac{a^3}{2}$ On BC, y = a $\therefore$ dy = 0 $\therefore \bar{F}.\, d\bar{r} = x^2 dx$ $\Rightarrow \int_{BC} \bar{F}.\, d\bar{r} = \int_a^0 x^2 dx = \left[\frac{x^3}{3}\right]_a^0 = \frac{-a^3}{3}$ On CO, x = 0 $\therefore \bar{F}.d\bar{r} = 0$ $\therefore \int_{CO} \bar{F}\,.\, d\bar{r} = 0$ From (1), we get $\int_C \bar{F}.\, d\bar{r} = \frac{a^3}{3} + \frac{a^3}{2} - \frac{a^3}{3} + 0 = \frac{a^3}{2}$. $\therefore \boxed{\int_C \bar{F}\,.\, d\,\bar{r} = \frac{a^3}{2}}$ **Ans.**	10	5	Low	Easy	CO-4	Numerical

6	Unit 6	Find the total Work done in above a particle in a field force given by F = 3xy $\hat{\imath}$ – 5z $\hat{\jmath}$+ 10x $\hat{k}$ along x = $t^2 + 1$, y = $2t^2$, z = t^3 from t = 1 to t = 2.	10	5	Low	Easy	CO-4	Numerical

		Soln. Here, $\bar{F} = 3xyi - 5zj + 10\,xk$ If $\bar{r}$ is the position vector at any point (x, y, z), then $\bar{r} = xi + yj + zk$ $\Rightarrow d\bar{r} = dxi + dyj + dzk$ Since, $x = t^2 + 1$, $dx = 2t\,dt$ $y = 2t^2$, $dy = 4t\,dt$ $z = t^3$, $dz = 3t^2\,dt$ $\therefore d\bar{r} = (2ti + 4tj + 3t^2 k)\,dt$ $\therefore \bar{F} \cdot d\bar{r} = [3(t^2+1)\,2t^2 i - 5t^3 j + 10(t^2+1)k] \cdot (2ti + 4tj + 3t^2 k)\,dt$ $= [6t^2(t^2+1)\,2t - 5t^3(4t) + 30t^2(t^2+1)]\,dt$ $= (12t^5 + 12t^3 - 20t^4 + 30t^4 + 30t^2)\,dt$ $= (12t^5 + 10t^4 + 12t^3 + 30t^2)\,dt$ $\therefore$ Required work done $= \int_{t=1}^{2} \bar{F} \cdot d\bar{r}$ $= \int_{t=1}^{2} (12t^5 + 10t^4 + 12t^3 + 30t^2)\,dt$ $= [2t^6 + 2t^5 + 3t^4 + 10t^3]_1^2$ $= 2(2^6 - 1) + 2(2^5 - 1) + 3(2^4 - 1) + 10(2^3 - 1)$ $= 2(63) + 2(31) + 3(15) + 10(7)$ $= 303$ $\therefore$ Work done = 303 **Ans.**						
7	Unit 6	Find work done by the variable force $\bar{F}$ = 2y $\hat{\imath}$ + xy$\hat{\jmath}$on a particle when it is displaced from the origin to the point$\bar{r} = 4\hat{\imath} + 2\hat{\jmath}$ along parabola $y^2 = x$.	10	5	Low	Easy	CO-4	Numerical

Soln.



Here, $x = y^2$

$\therefore \quad dx = 2y\,dy$

$\bar{r} = xi + yj$

$\therefore \quad d\bar{r} = d\,xi + d\,yj$

We have,

$\bar{F} = 2yi + xyj$

$\Rightarrow \bar{F}.d\bar{r} = 2y\,dx + xy\,dy$

$= 2y\ 2y\,dy + y^2 y\ dy \qquad [\because\ dx = 2ydy\ \&\ x = y^2]$

$= 4y^2 dy + y^3 dy$

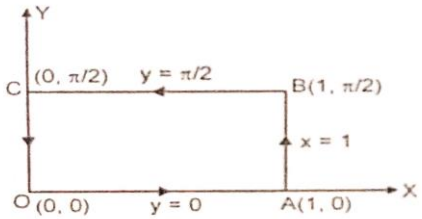
$= (y^3 + 4y^2)\ dy$

$\therefore$ Work done $= \int_C \bar{F}.d\bar{r}$

		$= \int_{y=0}^{2} (y^3 + 4y^2)\,dy$ $= \left[\frac{y^4}{4} + \frac{4}{3}y^3\right]_0^2$ $= 4 + \frac{32}{3}$ $= \frac{44}{3}$						
8	Unit 6	If $\bar{A} = (y - 2x)i + (3x + 2y)j$, find the circulation of $\bar{A}$ about a circle c in the XY-plane with center at origin and radius 2. C is traversed in the positive direction.	10	5	Low	Easy	CO-4	Numerical

		Soln. We have, $\overline{A} = (y-2x)\,i + (3x+2y)\,j$ The parametric equation of the circle $x^2+y^2=4$ is $x = 2\cos\theta$, $y = 2\sin\theta$, $z = 0$ and θ varies from 0 to 2π. Now, $\bar{r} = xi+yj+zk = 2\cos\theta\, i + 2\sin\theta\, j$ $\Rightarrow \frac{d\bar{r}}{d\theta} = -2\sin\theta\, i + 2\cos\theta\, j$ Also, $\overline{A} = (2\sin\theta - 4\cos\theta)\,i + (6\cos\theta + 4\sin\theta)\,j$ $\Rightarrow \overline{A}.\frac{d\bar{r}}{d\theta} = [(2\sin\theta - 4\cos\theta)\,i + (6\cos\theta + 4\sin\theta)\,j] \cdot [-2\sin\theta\, i + 2\cos\theta\, j]$ $\therefore \overline{A}.d\bar{r} = (-4\sin^2\theta + 16\sin\theta\cos\theta + 12\cos^2\theta)\,d\theta$ $\therefore$ Circulation of $\overline{A}$ about C $= \int_C \overline{A}.d\bar{r}$ $= \int_0^{2\pi} \overline{A}.\frac{d\bar{r}}{d\theta}\,d\theta$ $= \int_0^{2\pi} (-4\sin^2\theta + 16\sin\theta\cos\theta + 12\cos^2\theta)\,d\theta$ $= \int_0^{2\pi} \left[-4\left(\frac{1-\cos 2\theta}{2}\right) + 8\sin 2\theta + 12\left(\frac{1+\cos 2\theta}{2}\right)\right] d\theta$ $= \int_0^{2\pi} (4 + 8\cos 2\theta + 8\sin 2\theta)\,d\theta$ $= [4\theta + 4\sin 2\theta - 4\cos 2\theta]_0^{2\pi}$ $= 8\pi - 4 + 4$ $= 8\pi$ $\therefore$ **Circulation of $\overline{A}$ about C $= 8\pi$** **Ans.**						
9	Unit 6	If $\bar{F} = ie^x siny + je^x cosy$, evaluate $\int_c \bar{F}.\overline{dr}$, where c is the rectangle whose vertices are (0,0), (1,0), (1,$\frac{\pi}{2}$), (0,$\frac{\pi}{2}$).	10	5	Low	Easy	CO-4	Numerical

Soln.



Here, $\bar{F} = i\,e^x \sin y + j\,e^x \cos y$

In XY-plane the position vector at any point (x, y) is given by

$\bar{r} = xi + yj$

$\Rightarrow d\bar{r} = dx\,i + dy\,j$

$\therefore \bar{F} \cdot d\bar{r} = (i\,e^x \sin y + j\,e^x \cos y) \cdot (dx\,i + dy\,j)$

$= e^x \sin y\,dx + e^x \cos y\,dy$

$$\Rightarrow \int_C \bar{F} \cdot d\bar{r} = \int_{OA} \bar{F} \cdot d\bar{r} + \int_{AB} \bar{F} \cdot d\bar{r} + \int_{BC} \bar{F} \cdot d\bar{r} + \int_{CO} \bar{F} \cdot d\bar{r}$$

.....(1)

On OA, $y = 0 \Rightarrow dy = 0$

		$\Rightarrow \int_{OA} \bar{F} \cdot d\bar{r} = \int_0^1 0\,dx = 0$ $[\because \sin 0 = 0]$ On AB, $x = 1 \Rightarrow dx = 0$ $\Rightarrow \int_{AB} \bar{F} \cdot d\bar{r} = \int_0^{\pi/2} e \cos y\,dy = e[\sin y]_0^{\pi/2} = e$ On BC, $y = \frac{\pi}{2} \Rightarrow dy = 0$ $\int_{BC} \bar{F} \cdot d\bar{r} = \int_1^0 e^x\,dx = [e^x]_1^0 = 1 - e$ $[\because \sin \frac{\pi}{2} = 1]$ On CO, $x = 0 \Rightarrow dx = 0$ $\Rightarrow \int_{CO} \bar{F} \cdot d\bar{r} = \int_{\pi/2}^0 \cos y\,dy = [\sin y]_{\pi/2}^0 = 0 - 1 = -1$ From eqn. (1), $\int_C \bar{F} \cdot d\bar{r} = 0 + e + 1 - e - 1 = 0$ $\therefore \boxed{\int_C \bar{F} \cdot d\bar{r} = 0}$ **Ans.**						
10	Unit 6	Evaluate $\iint_s \bar{A} \cdot \hat{n}\,ds$; Where $\bar{A} = (x + y^2)\hat{\imath} - 2x\,\hat{\jmath} + 2yz\,\hat{k}$ and S is the surface of the plane 2x + y + 2z = 6 in the first	12	5	Medium	Moderate	CO-4	Numerical

octant.

Soln. A vector normal to the surface s is given by

$$\nabla(2x+y+2z) = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(2x+y+2z)$$

$$= 2i + j + 2k$$

$\therefore$ $\bar{n}$ = a unit vector normal to surface S

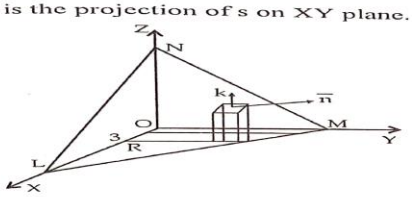
$$= \frac{2i+j+2k}{\sqrt{4+1+4}}$$

$$= \frac{2}{3}i + \frac{1}{3}j + \frac{2}{3}k$$

$$\Rightarrow \quad k.\bar{n} = k.\left(\frac{2}{3}i + \frac{1}{3}j + \frac{2}{3}k\right) = \frac{2}{3}$$

$$\therefore \quad \iint_S \bar{A}.\bar{n}\, ds = \iint_R \bar{A}.\bar{n}\, \frac{dx\, dy}{|k.\bar{n}|}$$

where R is the projection of s on XY plane.



Here, $\bar{A} = (x+y^2)i - 2xj + 2yzk$

$$\therefore \quad \bar{A}.\bar{n} = [(x+y^2)i - 2xj + 2yzk] \cdot \left(\frac{2}{3}i + \frac{1}{3}j + \frac{2}{3}k\right)$$

$$= \frac{2}{3}(x+y^2) - \frac{2}{3}x + \frac{4}{3}yz$$

$$= \frac{2}{3}y^2 + \frac{4}{3}yz$$

$$= \frac{2}{3}y^2 + \frac{4}{3}y\left(\frac{6-2x-y}{2}\right)$$

$$\left[\because \textit{On the plane } 2x+y+2z=6,\ z = \frac{6-2x-y}{2}\right]$$

$$= \frac{2}{3}y(y+6-2x-y)$$

$$= \frac{4}{3}y(3-x)$$

Hence $\iint_S \bar{A}.\bar{n}\, ds = \iint_R \bar{A}.\bar{n}\, \frac{dx\, dy}{|k.\bar{n}|}$

$$= \iint_R \frac{4}{3}y(3-x).\frac{3}{2}\, dx\, dy$$

$$= \int_0^3 \int_0^{6-2x} 2y(3-x)\, dy\, dx$$

		$= \int_0^3 2(3-x)\left[\frac{y^2}{2}\right]_0^{6-2x} dx$ $= \int_0^3 (3-x)(6-2x)^2 dx$ $= 4\int_0^3 (3-x)^3 dx$ $= 4\left[\frac{(3-x)^4}{4(-1)}\right]_0^3$ $= -(0-81)$ $= 81$ $\therefore \boxed{\iint_S \bar{A}.\bar{n}\, ds = 81}$ **Ans.**						
11	Unit 6	Evaluate $\iint_s \bar{A}.\hat{n}\, ds$; Where $\bar{A}$ = z$\hat{\imath}$ + x$\hat{\jmath}$ - 3y^2z $\hat{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between z=0 and z=5.	12	5	Medium	Moderate	CO-4	Numerical

Soln. A vector normal to the surface S is given by

$$\nabla(x^2+y^2)=\left(i\frac{\partial}{\partial x}+j\frac{\partial}{\partial y}+k\frac{\partial}{\partial z}\right)(x^2+y^2)$$

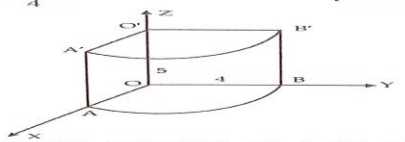
$$=2xi+2yj$$

$\therefore$ $\hat{n}$ = a unit vector normal to surface S

$$=\frac{2xi+2yj}{\sqrt{(2x)^2+(2y)^2}}$$

$$=\frac{xi+yj}{\sqrt{x^2+y^2}}$$

$$=\frac{xi+yj}{4} \quad [\because x^2+y^2=16]$$



Let R be the projection of S on YZ-plane, then

$$\iint_S \overline{A}\cdot\hat{n}\,dS=\iint_R \overline{A}\cdot\hat{n}\,\frac{dy\,dz}{|i\cdot\hat{n}|}$$

The region R is OB B' O' enclosed by $y=0$ to $y=4$ and $z=0$ to $z=5$

Now, $i\cdot\hat{n}=i\cdot\left(\frac{1}{4}xi+\frac{1}{4}yj\right)=\frac{1}{4}x$

Here, $\overline{A}=zi+xj-3y^2zk$

$$\therefore\ \overline{A}\cdot\hat{n}=(zi+xj-3y^2zk)\cdot\left(\frac{1}{4}xi+\frac{1}{4}yj\right)$$

$$=\frac{1}{4}zx+\frac{1}{4}xy$$

$$=\frac{1}{4}x(y+z)$$

$$\therefore\ \iint_S \overline{A}\cdot\hat{n}\,dS=\iint_R \overline{A}\cdot\hat{n}\,\frac{dy\,dz}{|i\cdot\hat{n}|}$$

$$=\iint_R \frac{1}{4}x(y+z)\,\frac{dy\,dz}{\frac{1}{4}x}$$

$$=\int_0^5\int_0^4 (y+z)\,dy\,dz$$

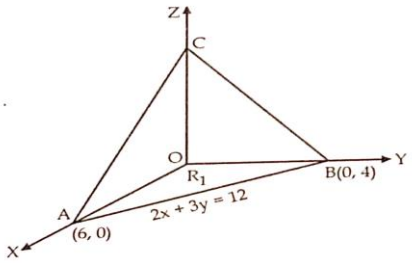
$$=\int_0^5\left[\frac{y^2}{2}+zy\right]_0^4 dz$$

$$=\int_0^5 (8+4z)\,dz$$

$$=[8z+2z^2]_0^5$$

$$=40+50=90$$

$\therefore$ $\boxed{\iint_S \overline{A}\cdot\hat{n}\,dS=90}$ Ans.

12	Unit 6	Find $\iint_S \bar{A}.\hat{n}\,ds$, Where $\bar{A} = 18\,z\,i - 12j + 3yk$ and S is the part of the plane 2x + 3y + 6z = 12 which is located in first octant. **Soln. Given :** $\overline{A} = 18\,z\,i - 12\,j + 3\,y\,k$ The equation of the plane is $\phi = 2x + 3y + 6z - 12$. Vector normal to the surface ϕ is ∇.  $\nabla\phi = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(2x + 3y + 6z - 12)$ $= 2i + 3j + 6k$ Now, $\hat{n}$ = unit normal $= \frac{2i + 3j + 6k}{\sqrt{4 + 9 + 36}} = \frac{1}{7}(2i + 3j + 6k)$ The projection of the plane 2x + 3y + 6z = 12 on the plane z = 0 is 2x + 3y = 12. i.e. the projection of the plane ABC on XY-plane is OAB.	12	5	Medium	Moderate	CO-4	Numerical

		$\therefore \iint_S \overline{A} \cdot \hat{n}\, dS = \iint_{R_1} \overline{A} \cdot \hat{n} \frac{dx\, dy}{\lvert \hat{n} \cdot k \rvert}$ .....(1) Now $\overline{A} \cdot \hat{n} = (18zi - 12j + 3yk) \cdot \frac{1}{7}(2i + 3j + 6k)$ $= \frac{1}{7}(36z - 36 + 18y)$ $= \frac{1}{7}\left[36\left(\frac{12 - 2x - 3y}{6}\right) - 36 + 18y\right]$ $\left[\because 2x + 3y + 6z = 12 \quad \therefore z = \frac{12 - 2x - 3y}{6}\right]$ $= \frac{1}{7}(36 - 12x)$ and $\hat{n} \cdot k = \frac{1}{7}(2i + 3j + 6k) \cdot k = \frac{6}{7}$ $\therefore$ Equation (1) becomes $\iint_S \overline{A} \cdot \hat{n}\, dS = \int_0^6 \left[\int_0^{\frac{12-2x}{3}} \frac{1}{7}(36 - 12x)\frac{7}{6} dy\right] dx$ $= \int_0^6 \frac{1}{6}[36y - 12xy]_0^{\frac{12-2x}{3}} dx$ $= \frac{1}{6}\int_0^6 \left[36\left(\frac{12-2x}{3}\right) - 12x\left(\frac{12-2x}{3}\right)\right] dx$ $= \int_0^6 \frac{1}{6}(144 - 24x - 48x + 8x^2) dx$ $= \int_0^6 \left(24 - 12x + \frac{4}{3}x^2\right) dx$ $= \left[24x - 6x^2 + \frac{4}{9}x^3\right]_0^6$ $= 144 - 216 + 96$ $= 24$ $\therefore \boxed{\iint_S \overline{A} \cdot \hat{n}\, dS = 24}$ **Ans.**						
13	Unit 6	Evaluate $\iint_s \bar{A} \cdot \hat{n}\, ds$; where $\bar{A} = y\hat{\imath} + 2x\hat{\jmath} - 2\hat{k}$ and S is the surface of the plane 2x + y = 6 in the first octant cut by the plane z = 4..	12	5	Medium	Moderate	CO-4	Numerical

		Soln. Let, $f(x,y,z) = 2x + y = 6$ be given surface S, then unit vector normal to S in outward direction is $\bar{n} = \frac{\nabla f}{\lvert \nabla f \rvert} = \frac{2i+j}{\sqrt{4+1}} = \frac{2i+j}{\sqrt{5}}$						

		Here, $\overline{\Lambda} = yi + 2xj - zk$ $\therefore \iint_S \overline{\Lambda}.\bar{n}\, ds = \iint_S (yi + 2xj - zk).\left(\frac{2i + j}{\sqrt{5}}\right) ds$ $= \iint_S \left(\frac{2y + 2x}{\sqrt{5}}\right) ds$ $= \iint_S \frac{(y + 6)}{\sqrt{5}} ds \quad [\because 2x + y = 6]$ Consider, projection R of surfaces S in YZ-plane. $\therefore ds = \frac{dy.dz}{\lvert \bar{n}.i \rvert}$ $= \frac{dy\ dz}{2/\sqrt{5}}$ $\therefore \iint_S \overline{\Lambda}.\bar{n}\, ds = \iint_S \frac{(y + 6)}{\sqrt{5}} \cdot \frac{dy\, dz}{2/\sqrt{5}}$ $= \iint_R \left(\frac{y + 6}{2}\right) dy.dz$ In YZ-plane, x = 0 and hence y = 6. $[\because 2x + y = 6]$ Thus, in first octant y varies from 0 to 6 and z varies from 0 to 4. $\therefore \iint_S \overline{\Lambda}.\bar{n}\, ds = \int_0^4 \left[\int_0^6 \frac{y + 6}{2} dy\right] dz$ $= \frac{1}{2}\int_0^4 \left[\frac{y^2}{2} + 6y\right]_0^6 dz$ $= \frac{1}{2}\int_0^4 \left(\frac{36}{2} + 36\right) dz$ $= 27\int_0^4 dz$ $= 27 \times 4 = 108.$ $\therefore \boxed{\iint_S \overline{\Lambda}.\hat{n}\, ds = 108}$ **Ans.**						
14	Unit 6	Evaluate surface integral $\iint_s \left(yz\hat{\imath} + xz\hat{\jmath} + xy\hat{k}\right).d\bar{s}$, where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ in the first quadrant.	12	5	Medium	Moderate	CO-4	Numerical

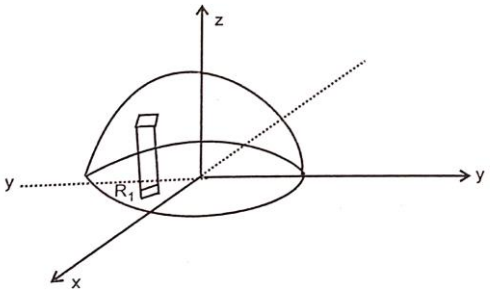
Soln. The equation of the sphere is

$$\phi = x^2 + y^2 + z^2 = 1$$

$\therefore$ Vector normal to the surface S is given by

$$\text{grad } \phi = \nabla\phi = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2)$$

$$= 2xi + 2yj + 2zk$$



		Also, $\hat{n}$ = a unit vector in the direction of $\nabla\phi$ $= \frac{2xi + 2yj + 2zk}{\sqrt{(2x)^2 + (2y)^2 + (2z)^2}}$ $= \frac{xi + yj + zk}{\sqrt{x^2 + y^2 + z^2}}$ $= xi + yj + zk$ $[\because x^2 + y^2 + z^2 = 1]$ The projection of the given surface (i.e. sphere) on XY-plane i.e. $z = 0$ is a circle $x^2 + y^2 = 1$. Now, $\iint_S \bar{F}.\bar{n}\, dS = \int_{R_1}\int \bar{F}.\bar{n}\, \frac{dx\, dy}{|\bar{n}.k|}$ ....(1) Here, $\bar{F} = yz\, i + zx\, j + xy\, k$ $\therefore \bar{F}.\bar{n} = (yzi + zxj + xyk).(xi + yj + zk)$ $= xyz + xyz + xyz = 3xyz$ & $\hat{n}.k = (xi + yj + zk).k = z$ $\therefore$ From equation (1), $\iint_S (yzi + zxj + xyk).\bar{n}\, dS = \int_0^1 \int_0^{\sqrt{1-y^2}} 3xy\, dx\, dy$ [$\because$ *Area is in first octant*] $= 3\int_0^1 \left[\frac{x^2}{2}\right]_0^{\sqrt{1-y^2}} y\, dy$ $= \frac{3}{2}\int_0^1 (1 - y^2)y\, dy$ $= \frac{3}{2}\int_0^1 (y - y^3)dy$ $= \frac{3}{2}\left[\frac{y^2}{2} - \frac{y^4}{4}\right]_0^1$ $= \frac{3}{2}\left[\frac{1}{2} - \frac{1}{4}\right]$ $= \frac{3}{8}$ $\therefore$ $\boxed{\iint_S (yzi + zxj + xyk).\bar{n}\, dS = \frac{3}{8}}$ **Ans.**						
15	Unit 6	Evaluate $\iint_s \bar{F}.\hat{n}\, ds$, where $\bar{F}$ = z $\hat{\imath}$ + x$\hat{\jmath}$-3y^2z$\hat{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between z = 0 and z = 5.	12	5	Medium	Moderate	CO-4	Numerical

Soln. A vector normal to the surface S is given by

$$\nabla(x^2+y^2)=\left(\hat{i}\frac{\partial}{\partial x}+\hat{j}\frac{\partial}{\partial y}+\hat{k}\frac{\partial}{\partial z}\right)(x^2+y^2)$$

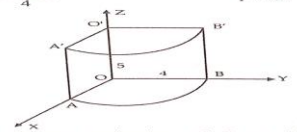
$$=2x\hat{i}+2y\hat{j}$$

$\therefore \hat{n}$ = a unit vector normal to surface S

$$=\frac{2x\hat{i}+2y\hat{j}}{\sqrt{(2x)^2+(2y)^2}}$$

$$=\frac{x\hat{i}+y\hat{j}}{\sqrt{x^2+y^2}}$$

$$=\frac{x\hat{i}+y\hat{j}}{4} \qquad [\because x^2+y^2=16]$$



Let R be the projection of S on YZ-plane, then

$$\iint_S \bar{A}\cdot\hat{n}\,dS=\iint_R \bar{A}\cdot\hat{n}\,\frac{dy\,dz}{|\hat{i}\cdot\hat{n}|}$$

The region R is OB B' O' enclosed by $y=0$ to $y=4$ and $z=0$ to $z=5$

Now, $\hat{i}\cdot\hat{n}=\hat{i}\cdot\left(\frac{1}{4}x\hat{i}+\frac{1}{4}y\hat{j}\right)=\frac{1}{4}x$

Here, $\bar{A}=z\hat{i}+x\hat{j}-3y^2z\hat{k}$

$$\therefore \bar{A}\cdot\hat{n}=(z\hat{i}+x\hat{j}-3y^2z\hat{k})\cdot\left(\frac{1}{4}x\hat{i}+\frac{1}{4}y\hat{j}\right)$$

$$=\frac{1}{4}zx+\frac{1}{4}xy$$

$$=\frac{1}{4}x(y+z)$$

$$\therefore \iint_S \bar{A}\cdot\hat{n}\,dS=\iint_R \bar{A}\cdot\hat{n}\,\frac{dy\,dz}{|\hat{i}\cdot\hat{n}|}$$

$$=\iint_R \frac{1}{4}x(y+z)\,\frac{dy\,dz}{\frac{1}{4}x}$$

$$=\int_0^5\int_0^4 (y+z)\,dy\,dz$$

$$=\int_0^5\left[\frac{y^2}{2}+zy\right]_0^4 dz$$

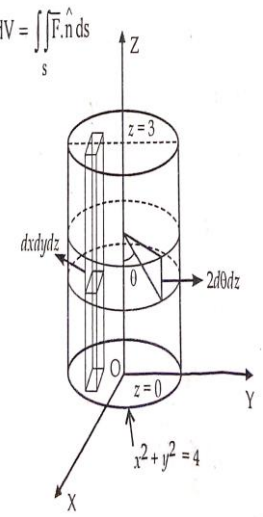
$$=\int_0^5 (8+4z)\,dz$$

$$=[8z+2z^2]_0^5$$

$$=40+50=90$$

$$\therefore \boxed{\iint_S \bar{A}\cdot\hat{n}\,dS=90}$$ Ans.

16	Unit 6	Using Gauss-divergence theorem evaluate $\iint_s \bar{F}.\hat{n}\ ds$, where S is the surface of the sphere $x^2 + y^2 + z^2 = 16$ *and* $\bar{F} = 3x\ i + 4y\ j + 5z\ k$ **Soln.** By Gauss's Divergence theorem, we have $\iint_S \bar{F}.\hat{n}\, dS = \iiint_V \nabla . \bar{F}\, dV$ Here, $\bar{F} = 3xi + 4yj + 5zk$ $\therefore \nabla.\bar{F} = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right).(3xi + 4yj + 5zk)$ $\therefore \nabla.\bar{F} = 3 + 4 + 5 = 12$ $\Rightarrow \iint_S \bar{F}.\hat{n}\, dS = \iiint_V 12\ dV$ $= 12 \iiint_V dV$ $= 12$ [Volume of sphere $x^2 + y^2 + z^2 = 16$] $= 12 \times \frac{4}{3}\pi(4)^3$ $\therefore \boxed{\iint_S \bar{F}.\hat{n}\, dS = 1024\pi}$ **Ans.**	12	5	Medium	Moderate	CO-4	Numerical
17	Unit 6	Evaluate $\iint_s \bar{F}.\hat{n}\ ds$, by using the divergence theorem for $\bar{F} = 4x\hat{\imath} - 2y^2\hat{\jmath} + z^2\hat{k}$taken over the region bounded by the cylinder $x^2 + y^2 = 4$; z = 0 and z = 3.	12	5	Medium	Moderate	CO-4	Numerical

		Soln. The Divergence theorem is $\iiint_V \nabla.\bar{F}\,dV = \iint_S \bar{F}.\hat{n}\,ds$  						

		Here, we have $\overline{F} = 4x\,i - 2y^2\,j + z^2\,k$ $\therefore \nabla.\overline{F} = \text{div}.\overline{F} = \frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2)$ $= 4 - 4y + 2z$ $\therefore \iiint_V \nabla.\overline{F}\,dV = \iiint_V (4 - 4y + 2z)dV$ $= \iiint_V (4 - 4y + 2z)\,dxdydz$ To cover the whole volume of cylinder $x^2 + y^2 = 4$, $z = 0$, $z = 3$. 'z' varies from 0 to 3, y varies from $-\sqrt{4-x^2}$ to $\sqrt{4-x^2}$ and x varies from –2 to 2. $\therefore \iiint_V \nabla.\overline{F}\,dV$ $= \int_{x=-2}^{2} \int_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{z=0}^{3} (4 - 4y + 2z)\,dx\,dy\,dz$ $= \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \left[4z - 4yz + z^2\right]_0^3 dxdy$ $= \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (12 - 12y + 9)\,dxdy$ $= \int_{-2}^{2} \left[\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (21 - 12y)\,dy\right] dx$ $= \int_{-2}^{2} \left[2\int_{0}^{\sqrt{4-x^2}} 21\,dy\right] dx$ *[$\because$ 12y is odd and $\int_{-a}^{a} f(x)\,dx = 0$, if f(x) is odd.]* $= \int_{-2}^{2} 42\,[y]_0^{\sqrt{4-x^2}}\,dx$ $= 42 \int_{-2}^{2} \sqrt{4-x^2}\,dx$ $= 42 \times 2 \int_{0}^{2} \sqrt{4-x^2}\,dx$ *[$\because$ $\sqrt{4-x^2}$ is even function]* $= 84\left[\frac{x}{2}\sqrt{4-x^2} + \frac{4}{2}\sin^{-1}\frac{x}{2}\right]_0^2$ $= 84\pi$						
18	Unit 6	State and use divergence theorem to evaluate $\iint_S \bar{F}.\hat{n}\,ds$,	12	5	Medium	Moderate	CO-4	Numerical

		where $\bar{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and s is the surface of the cube bounded by x=0, x=1, y=0,y=1, z=0, z=1. Soln. By Divergence theorem, we have $\iint_S \bar{F}.\hat{n}\, dS = \iiint_V (\nabla.\bar{F})\, dV$ $= \iiint_V \left[\frac{\partial}{\partial x}(4xz) + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(yz)\right] dx\, dy\, dz$ $= \iiint_V (4z - 2y + y)\, dx\, dy\, dz$ $= \iiint_V (4z - y)\, dx\, dy\, dz$ $= \int_0^1 \int_0^1 \int_0^1 (4z - y)\, dx\, dy\, dz$ $= \int_0^1 \int_0^1 \left(\frac{4z^2}{2} - yz\right)_0^1 dx\, dy$						

		$= \int_0^1 \int_0^1 \left(2z^2 - yz\right)_0^1 dx\ dy$ $= \int_0^1 \int_0^1 (2 - y) dx\ dy$ $= \int_0^1 \left(2y - \frac{y^2}{2}\right)_0^1 dx$ $= \frac{3}{2} \int_0^1 dx$ $= \frac{3}{2}(x)_0^1$ $= \frac{3}{2}$ $\therefore \boxed{\iint_S \overline{F} \cdot \hat{n}\ dS = \frac{3}{2}}$ **Ans.**						
19	Unit 6	Evaluate $\iint_s \ \bar{F} \cdot \hat{n}\ ds$ over the entire surface of the region above the xy-plane bounded by the cone $z^2 = x^2 + y^2$ and the plane z=4; if $\bar{F} = 4xz\hat{i} + xyz^2\hat{j} + 3z\hat{k}$.	12	5	Medium	Difficult	CO-4	Numerical

Soln. **Given:** $\overline{F} = 4xz\,\hat{i} + xyz^2\,\hat{j} + 3z\hat{k}$

If V is the volume enclosed by s, then V is bounded by the surface $z = 0$, $z = 4$, $z^2 = x^2 + y^2$.

By Divergence theorem, we have

$$\iint_S \overline{F}\cdot\hat{n}\,ds = \iiint_V \operatorname{div}\overline{F}\,dV$$

$$= \iiint_V \left[\frac{\partial}{\partial x}(4xz) + \frac{\partial}{\partial y}(xyz^2) + \frac{\partial}{\partial z}(3z)\right] dV$$

$$= \iiint_V (4z + xz^2 + 3)\,dV$$

$$= \int_0^4 \int_{-z}^{z} \int_{-\sqrt{z^2-y^2}}^{\sqrt{z^2-y^2}} (4z + xz^2 + 3)\,dx\,dy\,dz$$

$$= 2\int_0^4 \int_{-z}^{z} \int_0^{\sqrt{z^2-y^2}} (4z+3)\,dx\,dy\,dz$$

$$\left[\because \int_{-\sqrt{z^2-y^2}}^{\sqrt{z^2-y^2}} x\,dx = 0\right]$$

$$= 2\int_0^4 \int_{-z}^{z} (4z+3)\sqrt{z^2 - y^2}\,dy\,dz$$

$$= 4\int_0^4\int_0^z (4z+3)\sqrt{z^2-y^2}\,dy\,dz$$

$$\left[\because \sqrt{z^2-y^2} \text{ is an even function}\right]$$

$$= 4\int_0^4 (4z+3)\left[\int_0^z \sqrt{z^2-y^2}\,dy\right]dz$$

$$= 4\int_0^4 (4z+3)\left[\frac{y\sqrt{z^2-y^2}}{2} + \frac{z^2}{2}\sin^{-1}\frac{y}{z}\right]_0^z dz$$

$$\left[\because \int\sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right]$$

$$= 4\int_0^4 (4z+3)\left[\frac{z^2}{2}\sin^{-1}1\right]dz$$

$$= 4\int_0^4 (4z^3+3z^2)\left(\frac{1}{2}\frac{\pi}{2}\right)dz$$

$$= 4\times\frac{\pi}{4}\int_0^4 (4z^3+3z^2)\,dz$$

$$= \pi[z^4+z^3]_0^4$$

$$= \pi[256+64]$$

$$= 320\,\pi$$

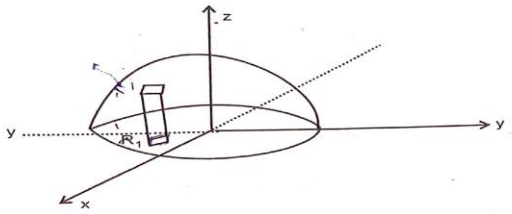
$$\therefore \boxed{\iint_S \bar{F}\cdot\hat{n}\,.\,ds = 320\,\pi} \quad \textbf{Ans.}$$

20	Unit 6	Evaluate $\iint_s (y^2z^2)\hat{i} + (z^2x^2)\hat{j} + (z^2y^2)\hat{k}.\hat{n}ds$, where s is the part of the sphere $x^2 + y^2 + z^2 = 1$, above the xy plane and bounded by this plane. **Soln.** The equation of the sphere is $\phi = x^2 + y^2 + z^2 = 1$ $\therefore$ Vector normal to the surface S is given by grad $\phi = \nabla\phi$ $= \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(x^2 + y^2 + z^2)$ $= 2xi + 2yj + 2zk$ Also, $\hat{n}$ = unit vector in the direction of $\nabla\phi$. $= \frac{\nabla\phi}{\lvert\nabla\phi\rvert}$ $= \frac{2xi + 2yj + 2zk}{\sqrt{4x^2 + 4y^2 + 4z^2}}$ $= \frac{xi + yj + zk}{\sqrt{x^2 + y^2 + z^2}}$ $= xi + yj + zk \quad [\because x^2 + y^2 + z^2 = 1]$ The projection of the given surface (i.e. sphere) on XY-plane is circle $x^2 + y^2 = 1$. Now, we know that	12	5	Medium	Difficult	CO-4	Numerical

$$\iint_S \bar{F} \cdot d\bar{S} = \iint_S \bar{F} \cdot \hat{n}\, dS = \iint_{R_1} \bar{F} \cdot \hat{n}\, \frac{dx\, dy}{|\hat{n} \cdot k|}$$

$$\therefore \iint_S (y^2 z^2 i + z^2 x^2 j + x^2 y^2 k) \cdot dS$$

$$= \iint_{R_1} (y^2 z^2 i + z^2 x^2 j + x^2 y^2 k) \cdot \hat{n}\, \frac{dx\, dy}{|\hat{n} \cdot k|} \quad \ldots.(1)$$



where R_1 is the projection of S (i.e. sphere) on the XY-plane which is circle $x^2 + y^2 = 1$. Obviously in R_1, x varies from -1 to 1 and y varies from $-\sqrt{1-x^2}$ to $\sqrt{1-x^2}$

$$[\because\ x^2 + y^2 = 1 \Rightarrow y = \pm\sqrt{1-x^2}\ ]$$

Now, $(y^2 z^2 i + z^2 x^2 j + x^2 y^2 k) \cdot \hat{n}$

$$= (y^2 z^2 i + z^2 x^2 j + x^2 y^2 k) \cdot (xi + yj + zk)$$

$$= xy^2 z^2 + yz^2 x^2 + zx^2 y^2$$

and $\hat{n} \cdot k = (xi + yj + zk) \cdot k = z$

$\therefore$ Equation (1) becomes

$$\iint_S (y^2 z^2 i + z^2 x^2 j + x^2 y^2 k) \cdot dS$$

$$= \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (xy^2 z + yzx^2 + x^2 y^2)\, dx\, dy$$

$$= \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} [(xy^2 + yx^2) z + x^2 y^2]\, dx\, dy$$

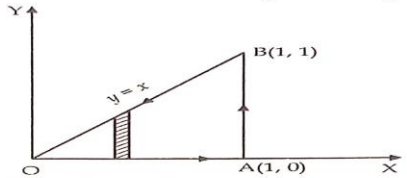
$$= \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} [(xy^2 + yx^2)\sqrt{1-x^2-y^2} + x^2 y^2]\, dx dy$$

$$[\because\ x^2 + y^2 + z^2 = 1 \Rightarrow z = \sqrt{1-x^2-y^2}\ ]$$

		$= 2.2 \int_0^1 \int_0^{\sqrt{1-x^2}} x^2 y^2 \, dx \, dy$ *[∵ $(xy^2 + yx^2)\sqrt{1-x^2-y^2}$ is an odd function of x and y & x^2y^2 is an even function of x & y]* $= 4 \int_0^1 x^2 \left[\frac{y^3}{3}\right]_0^{\sqrt{1-x^2}} dx$ $= \frac{4}{3} \int_0^1 x^2 (1-x^2)^{3/2} dx$ Put $x = \sin 0$ so that $dx = \cos 0 \, d0$ as $x \to 0$, $0 \to 0$ and $x \to 1$, $0 \to \frac{\pi}{2}$ $= \frac{4}{3} \int_0^{\pi/2} \sin^2 0 \cos^4 0 \, d0$ $= \frac{4}{3} \frac{1.3.1}{6.4.2} \frac{\pi}{2}$ $= \frac{\pi}{24}$ *[∵ By prop. of Beta function]* ∴ $\boxed{\iint_S (y^2 z^2 i + z^2 x^2 j + x^2 y^2 k).dS = \frac{\pi}{24}}$ **Ans.**						
21	Unit 6	Evaluate $\oint_c \bar{F}.\, d\bar{r}$ by Stoke's theorem , where $\bar{F} = y^2\hat{i} + x^2\hat{j} - (x+z)\hat{k}$ and c is the boundary of the triangle with vertices at (0,0,0), (1,0,0) and (1,1,0).	12	5	Medium	Difficult	CO-4	Numerical

Soln. Stoke's Theorem :

Statement : If S is an open surface bounded by a simple closed curve C and $\overline{A}$ is a vector field with continuous first order partial derivatives, then $\int_C \overline{A} \cdot d\overline{r} = \iint_S (\nabla \times \overline{A}) \cdot \overline{n}\, ds$.



Given : $\overline{F} = y^2\,i + x^2\,j - (x+z)\,k$

Since, z co-ordinates of each vertex of the triangle is zero, therefore the triangle lies in XY-plane and $\overline{n} = k$.

$$\therefore \text{Curl } \overline{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & -(x+z) \end{vmatrix}$$

$$= i(0-0) - j(-1-0) + k(2x-2y)$$

$$= \overline{j} + 2(x-y)\overline{k}$$

$$\Rightarrow \text{Curl } \overline{F}.\overline{n} = [\overline{j} + 2(x-y)\overline{k}].[\overline{k}]$$

$$= 2(x-y)$$

$$\text{and } ds = \frac{dx\,dy}{|\overline{n} \cdot k|} = \frac{dx\,dy}{|k \cdot k|} = dx\,dy$$

The equation of line OB is $y = x$.

$\therefore$ By Stoke's theorem,

$$\int_C \overline{F}.d\overline{r} = \iint_S \text{curl}\,\overline{F} \,.\, \overline{n}\, ds$$

$$= \int_0^1 \int_0^x 2(x-y)\,dy\,dx$$

$$= \int_0^1 2\left[xy - \frac{y^2}{2}\right]_0^x dx$$

$$= 2\int_0^1 \left(x^2 - \frac{x^2}{2}\right) dx$$

		$= \int_0^1 x^2\, dx$ $= \left[\frac{x^3}{3}\right]_0^1$ $= \frac{1}{3}$. $\therefore \boxed{\int_C \bar{F} \cdot dr = \frac{1}{3}}$ **Ans.**						
22	Unit 6	State divergence theorem and hence use it to evaluate $\iint_s \bar{F}.\hat{n}\, ds$, for $\bar{F} = (x^2 - yz)\hat{i} + (z^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$taken over the rectangular parallelopiped 0≤x≤a, 0≤y≤b, 0≤z≤c.	12	5	Medium	Difficult	CO-4	Numerical

Soln. We have, $\overline{F} = (x^2 - yz)i + (y^2 - zx)j + (z^2 - xy)k$

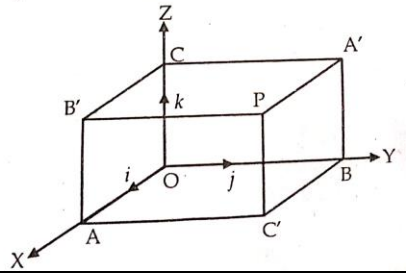
For verification of Divergence theorem, we shall evaluate the volume and surface integrals separately and showing that they are equal.

Now,

$$\text{div } \overline{F} = \frac{\partial}{\partial x}(x^2 - yz) + \frac{\partial}{\partial y}(y^2 - zx) + \frac{\partial}{\partial z}(z^2 - yx)$$

$$= 2(x + y + z)$$

$$\therefore \quad \iiint_V \text{div } \overline{F}\, dV = \int_0^c \int_0^b \int_0^a 2(x + y + z)\, dx\, dy\, dz$$



$$= \int_0^c \int_0^b 2 \left[\frac{x^2}{2} + yx + zx \right]_0^a dy$$

$$= \int_0^c \int_0^b 2 \left(\frac{a^2}{2} + ya + za \right) dy\ dz$$

$$= \int_0^c 2 \left[\frac{a^2}{2} y + \frac{ay^2}{2} + azy \right]_0^b dz$$

$$= 2 \int_0^c \left(\frac{a^2 b}{2} + \frac{ab^2}{2} + abz \right) dz$$

$$= 2 \left[\frac{a^2 b}{2} z + \frac{ab^2}{2} z + ab \frac{z^2}{2} \right]_0^c$$

$$= a^2 bc + ab^2 c + abc^2$$

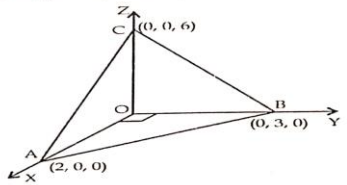
$$= abc(a + b + c)$$

23	Unit 6	Apply stokes theorem to evaluate $\int_c [(x+y)dx + (2x-z)dy + (y+z)dz]$, where c is the boundary of the triangle with vertices (2, 0, 0), (0,3,0) and (0,0,6).	12	5	Medium	Difficult	CO-4	Numerical

Soln. We have to evaluate

$$\int_C [(x+y)\,dx + (2x-z)\,dy + (y+z)\,dz]$$

$\therefore \overline{F} = (x+y)\,i + (2x-z)\,j + (y+z)\,k$



$\therefore$ By Stoke's theorem,

$$\int_C \overline{F}.\,d\overline{r} = \iint_S (\nabla \times \overline{F}).\,\hat{n}\,ds$$

Now,

$$\nabla \times \overline{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y & 2x-z & y+z \end{vmatrix}$$

$$= i(1+1) - j(0-0) + k(2-1)$$

$\therefore \nabla \times \overline{F} = 2i + k$

C is the boundary of the triangle with vertices (2, 0, 0), (0, 3, 0), (0, 0, 6).

$\therefore \frac{x}{2} + \frac{y}{3} + \frac{z}{6} = 1$

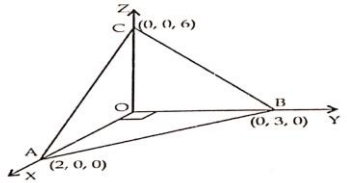
$\Rightarrow \frac{3x+2y}{6} + \frac{z}{6} = 1$

$\Rightarrow 3x + 2y + z = 6$

$\therefore \phi = 3x + 2y + z - 6$

		$\therefore \nabla\phi = \frac{\partial\phi}{\partial x}\mathrm{i} + \frac{\partial\phi}{\partial y}\mathrm{j} + \frac{\partial\phi}{\partial z}\mathrm{k}$ $\therefore \nabla\phi = 3\mathrm{i} + 2\mathrm{j} + \mathrm{k}$ $\hat{n} = \frac{\nabla\phi}{\lvert\nabla\phi\rvert} = \frac{3\mathrm{i}+2\mathrm{j}+\mathrm{k}}{\sqrt{9+4+1}} = \frac{3\mathrm{i}+2\mathrm{j}+\mathrm{k}}{\sqrt{14}}$ and $\hat{n}\,.\,\mathrm{k} = 1/\sqrt{14}$ $\therefore \iint_S (\nabla\times\bar{F}).\hat{n}\,ds = \iint_S (2\mathrm{i}+\mathrm{k}).\left(\frac{3\mathrm{i}+2\mathrm{j}+\mathrm{k}}{\sqrt{14}}\right)\frac{dx\,dy}{\lvert\hat{n}\,.\,\mathrm{k}\rvert}$ $= \iint_S \left(\frac{6+1}{\sqrt{14}}\right)\frac{dx\,dy}{(1/\sqrt{14})}$ $= \iint_S 7\,dx\,dy$ $= 7\iint_S dx\,dy$ $= 7$ (Area of Δ OAB) Now Area of Δ OAB $= \frac{1}{2}$ base$\times$ height $= \frac{1}{2}\times 2\times 3$ $= 3$ $\therefore \iint_S (\nabla\times\bar{F}).\bar{n}\,ds = 7\times 3 = 21$ $\therefore \int_C \bar{F}.d\bar{r} = \iint_S (\nabla\times F).\bar{n}\,ds$ *[∵ By Stoke's theorem]* $\therefore \boxed{\int_C [(x+y)\,dx + (2x-z)\,dy + (y+z)\,dz] = 21}$ **Ans.**						
24	Unit 6	Apply stokes theoremEvaluate $\oint_c \bar{F}.\,d\bar{r}$ for vector field $\bar{F} = (x^2 - y^2)\hat{\imath} + 2xy\hat{\jmath}$ in the rectangle in xy-plane	15	5	High	Very difficult	CO-4	Numerical

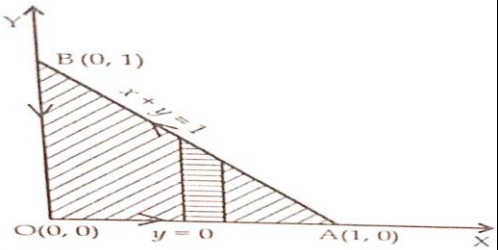
		bounded by lines x=0 , x=b, y=0, y=b. We have $\nabla \times \overline{F} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 - y^2 & 2xy & 0 \end{vmatrix}$ $= (2y + 2y)k = 4yk$ Also the unit vector normal to the surface s in outward direction is $\overline{n} = k$. *[∵ Z-axis is normal to surfaces]* Also in XY-plane $ds = \frac{dxdy}{\lvert \hat{n} \cdot k \rvert} = \frac{dxdy}{\lvert k \cdot k \rvert} = dxdy$ $\therefore \iint_S (\nabla \times \overline{F})\, \overline{n}\, ds = \iint_R 4yk.k\, dx\, dy = \iint_R 4y\, dx\, dy.$ where R be the region of the surface s. Consider a strip parallel to Y-axis. This strip starts on line y = 0 (i.e. X-axis) and end on the line y = b. We move this strip from x = 0 (Y-axis) to x = a to cover complete region R. $\therefore \iint_S (\nabla \times \overline{F}).\overline{n}\, ds = \int_0^a \left[\int_0^b 4ydy\right] dx$ $= \int_0^a \left[2y^2\right]_0^b dx$ $= \int_0^a 2b^2 dx$ $= 2b^2[x]_0^a$ $= 2ab^2$ .....(3) ∴ From eqn. (2) and (3), we get $\int_C \overline{F}.\, d\overline{r} = \iint_S (\nabla \times \overline{F}).\overline{n}\, ds$ and hence the Stoke's theorem is verified. **Ans.**							
25	Unit 6	Apply Stoke's theorem to evaluate	15	5	High	Very difficult	CO-4	Numerical

		$\oint_C [(x+y)dx + (2x-z)dty + (y+z)dz]$ where C is the boundary of the triangle with vertices (2,0,0) , (0, 3, 0) & (0,0,6). **Soln.** We have to evaluate $\int_C [(x+y)\,dx + (2x-z)\,dy + (y+z)\,dz]$ $\therefore \ \overline{F} = (x+y)\,i + (2x-z)\,j + (y+z)\,k$  $\therefore$ By Stoke's theorem, $\int_C \overline{F}.\,d\overline{r} = \iint_S (\nabla \times \overline{F}).\,\hat{n}\,ds$ Now, $\nabla \times \overline{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y & 2x-z & y+z \end{vmatrix}$ $= i\,(1+1) - j\,(0-0) + k\,(2-1)$ $\therefore \ \nabla \times \overline{F} = 2i + k$ C is the boundary of the triangle with vertices (2, 0, 0), (0, 3, 0), (0, 0, 6). $\therefore \ \frac{x}{2} + \frac{y}{3} + \frac{z}{6} = 1$ $\Rightarrow \frac{3x+2y}{6} + \frac{z}{6} = 1$ $\Rightarrow 3x + 2y + z = 6$ $\therefore \ \phi = 3x + 2y + z - 6$						

		$\therefore \nabla\phi = \frac{\partial\phi}{\partial x}\mathrm{i} + \frac{\partial\phi}{\partial y}\mathrm{j} + \frac{\partial\phi}{\partial z}\mathrm{k}$ $\therefore \nabla\phi = 3\mathrm{i} + 2\mathrm{j} + \mathrm{k}$ $\hat{n} = \frac{\nabla\phi}{\lvert\nabla\phi\rvert} = \frac{3\mathrm{i}+2\mathrm{j}+\mathrm{k}}{\sqrt{9+4+1}} = \frac{3\mathrm{i}+2\mathrm{j}+\mathrm{k}}{\sqrt{14}}$ and $\hat{n}.\mathrm{k} = 1/\sqrt{14}$ $\therefore \iint_S (\nabla\times\overline{F}).\hat{n}\,ds = \iint_S (2\mathrm{i}+\mathrm{k}).\left(\frac{3\mathrm{i}+2\mathrm{j}+\mathrm{k}}{\sqrt{14}}\right)\frac{dx\,dy}{\lvert\hat{n}.\mathrm{k}\rvert}$ $= \iint_S \left(\frac{6+1}{\sqrt{14}}\right)\frac{dx\,dy}{(1/\sqrt{14})}$ $= \iint_S 7\,dx\,dy$ $= 7\iint_S dx\,dy$ $= 7$ (Area of Δ OAB) Now Area of Δ OAB $= \frac{1}{2}$ base$\times$ height $= \frac{1}{2}\times 2\times 3$ $= 3$ $\therefore \iint_S (\nabla\times\overline{F}).\overline{n}\,ds = 7\times 3 = 21$ $\therefore \int_C \overline{F}.d\overline{r} = \iint_S (\nabla\times F).\overline{n}\,ds$ *[$\because$ By Stoke's theorem]* $\therefore \boxed{\int_C [(x+y)\,dx + (2x-z)\,dy + (y+z)\,dz] = 21}$ **Ans.**						
26	Unit 6	State Greens theorem and Evaluate in the plane for $\oint(3x^2 - 8y^2)dx + (4y - 6xy)dy$,where c is boundary of the region x=0, y=0, x+y=1.	15	5	High	Very difficult	CO-4	Numerical

Soln. Here, $M = 3x^2 - 8y^2$,

$N = 4y - 6xy$



$$\Rightarrow \frac{\partial M}{\partial y} = -16y,$$

$$\frac{\partial N}{\partial x} = -6y$$

$$\Rightarrow \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 10y$$

To Green's theorem, we have to show that

$$\int_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

Now,

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

$$= \int_{x=0}^{1} \int_{y=0}^{1-x} 10y\,dx\,dy$$

$$= \int_{x=0}^{1} \left[\int_{y=0}^{1-x} 10y\,dy\right] dx$$

$$= \int_0^1 5\left[y^2\right]_0^{1-x}\,dx$$

$$= 5\int_0^1 (1-x)^2\,dx$$

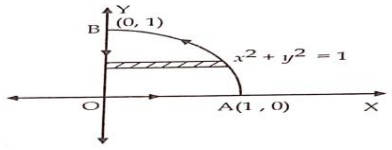
$$= 5\int_0^1 (1 - 2x + x^2)\,dx$$

$$= 5\left[x - x^2 + \frac{x^3}{3}\right]_0^1$$

$$\therefore \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right] dx\,dy = \frac{5}{3}$$

27	Unit 6	Use Green's theorem in a plane to evaluate the integral $\oint_c (2x^2 - y^2)dx + (x^2 + y^2)dy$, Where c is the boundary in xy-plane of the area enclosed by the x-axis and the semi-circle$x^2 + y^2 = 1$ in the upper half xy-plane.	15	5	High	Very difficult	CO-4	Numerical

Soln.



Here, $M = 2x^2 - y^2$, $N = x^2 + y^2$

$\frac{\partial M}{\partial y} = -2y, \frac{\partial N}{\partial x} = 2x$

$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2x + 2y = 2(x + y)$

By Green's theorem,

$$\int_C M dx + N dy = \int\int_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\, dy$$

$$= \iint_R 2(x + y)\, dx\, dy$$

$$= \int_{y=0}^{1} \int_{x=0}^{\sqrt{1-y^2}} 2(x + y)\, dx\, dy$$

$$= 2\int_{y=0}^{1} \left[\frac{x^2}{2} + xy\right]_0^{\sqrt{1-y^2}} dy$$

$$= \int_{y=0}^{1} \left[(1 - y^2) + 2y\sqrt{1 - y^2}\right] dy$$

		$= \left[y - \frac{y^3}{3}\right]_0^1 + 2\int_{y=0}^{1} y\sqrt{1-y^2}\,dy$ Put $y = \sin\theta$, $dy = \cos\theta\, d\theta$. When $y = 0$, $\theta = 0$ and $y = 1$, $\theta = \pi/2$ $= 1 - \frac{1}{3} + 2\int_0^{\pi/2} \sin\theta.\cos\theta.\cos\theta.\, d\theta$ $= \frac{2}{3} + 2\int_0^{\pi/2} \sin\theta \cdot \cos^2\theta\, d\theta$ $= \frac{2}{3} + 2\left[\frac{\cos^3\theta}{-3}\right]_0^{\pi/2}$ $= \frac{2}{3} + \frac{2}{3}$ $= \frac{4}{3}$ $\therefore \boxed{\int_C [Mdx + Ndy] = \frac{4}{3}}$ **Ans.**						
28	Unit 6	ApplyStoke's theorem evaluate $\oint \bar{F} . \overline{dr}$ for the vector field $\bar{F} = (2x - y)\hat{\imath} - yz^2\hat{\jmath} - y^2z\hat{k}$ over the upper half surface $x^2 + y^2 + z^2 = 1$bounded by it's projection on the xy-plane. $\text{Curl } \bar{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2z \end{vmatrix}$ $= (-2yz + 2yz)i + (0 - 0)j + (0 + 1)k = k$ In xy plane $\hat{n} = k$ $\therefore \text{Curl } \bar{F}.\bar{n} = k.\bar{n} = k.k = 1$	15	5	High	Very difficult	CO-4	Numerical

		$\therefore \iint_S \text{curl } \bar{F} \cdot \bar{n}\, ds = \iint_S (1) \frac{dxdy}{\lvert \bar{n} \cdot \bar{k} \rvert} = \iint_S dxdy$ where R is the projection of S on XY-plane. $= \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy\, dx$ $= \int_{-1}^{1} 2\sqrt{1-x^2}\, dx$ $= 4\int_{0}^{1} \sqrt{1-x^2}\, dx$ *[$\because \sqrt{1-x^2}$ is even]* $= 4\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1} x\right]_0^1$ $= 4\left[\frac{1}{2} \cdot \frac{\pi}{2}\right]$ $= \pi$						
29	Unit 6	Apply stokes theorem to evaluate $\oint_c [sinzdx - cosxdy + sinydz]$,where c is the boundary of the rectangle $0 \le x \le \pi, 0 \le y \le 1$, $z=3$ oriented clockwise as viewed from the origin.	15	5	High	Very difficult	CO-4	Numerical

Soln. Consider, $\int_C (\sin z\,dx - \cos x\,dy + \sin y\,dz)$

$$= \int_C [\sin z\,i - \cos x\,j + \sin y\,k].(dx\,i + dy\,j + dz\,k)$$

$$= \int_C [(\sin z)\,i - (\cos x)\,j + (\sin y)\,k].\,d\bar{r}$$

$$= \iint_S \text{curl}\,\{(\sin z)\,i - (\cos x)\,j + (\sin y)\,k\}.\,\hat{n}\,ds$$

....(1)

$$= \iint_S (\nabla \times \bar{F}).\,\hat{n}\,ds \qquad [\because \textit{By Stoke's theorem}]$$

where $\bar{F} = \sin z\,i - \cos x\,j + \sin y\,k$

$$\therefore \quad \nabla \times \bar{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sin z & -\cos x & \sin y \end{vmatrix}$$

$$= i(\cos y - 0) - j(0 - \cos z) + k(\sin x - 0)$$

$$= (\cos y)\,i + (\cos z)\,j + (\sin x)\,k$$

On XY-plane $\hat{n} = k$

$$\Rightarrow (\nabla \times \bar{F}).\,\hat{n} = [\cos y\,i + \cos z\,j + \sin x\,k].k = \sin x$$

$$\& \quad ds = \frac{dx\,dy}{|\hat{n}.k|} = \frac{dx\,dy}{|k.k|} = dx\,dy$$

$$\therefore \quad \text{From (1)}, \int_C (\sin z\,dx - \cos x\,dy + \sin y\,dz)$$

$$= \int_0^{\pi} \int_0^{1} \sin x\,dx\,dy$$

$$= \int_0^{\pi} \sin x\,[y]_0^1\,dx$$

$$= \int_0^{\pi} \sin x\,dx$$

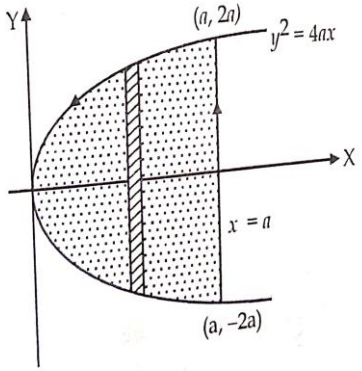
$$= [-\cos x]_0^{\pi}$$

$$= -[\cos \pi - \cos 0]$$

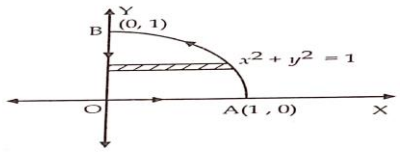
$$= -[-1 - 1]$$

$$= 2$$

$$\therefore \quad \boxed{\int_C (\sin z\,dx - \cos x\,dy + \sin y\,dz) = 2} \quad \textbf{Ans.}$$

30	Unit 6	Use Green's theorem to evaluate $\int_c \bar{F} \cdot \overline{dr}$, where $\bar{F} = (2xy^2 + 3z^2)\hat{i} + (x^2 + 4yz)\hat{j} + (2y^2 + 6xz)\hat{k}$and c is the curve enclosing a region $y^2 = 4ax$; x = a in the plane z = 0. Soln. On XY-plane $\bar{r} = xi + yj$ $\therefore$ $d\bar{r} = dxi + dyj$  Here, we have	15	5	High	Very difficult	CO-4	Numerical

		$\bar{F} = (2xy^2 + 3z^2)\,\hat{i} + (x^2 + 4yz)\,\hat{j} + (2y^2 + 6xz)\,\hat{k}$ $\therefore \bar{F}.d\bar{r} = [(2xy^2 + 3z^2)\,i + (x^2 + 4yz)\,j + (2y^2 + 6xz)\,k].(dx\,i + dy\,j)$ $= 2xy^2dx + x^2dy$ *[∵ z = 0 hence, dz = 0]* $\Rightarrow \int_C \bar{F}.d\bar{r} = \int_C (2xy^2dx + x^2dy)$ By Green's theorem, we have $\int_C \bar{F}.d\bar{r} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy,$ where $M = 2xy^2, N = x^2$. $= \int_{x=0}^{a} \int_{y=-2a}^{2a} (2x - 4xy)dx\,dy$ $= 2\int_0^a \int_{y=0}^{2a} 2x\,dx\,dy$ *[∵ 4xy is odd w.r.to y and $\int_{-a}^{a} f(x)dx = 0$, if $f(x)$ is odd]* $= 4\int_0^a x\,[y]_0^{2a}\,dx$ $= 8a\int_0^a x\,dx$ $= 8a\left(\frac{x^2}{2}\right)_0^a$ $= 4a^3$ $\therefore \boxed{\int_C \bar{F}.d\bar{r} = 4a^3}$ **Ans.**						
31	Unit 6	Use Green theorem in a plane to evaluate the integral $\int_c (2x^2 - y^2)dx + (x^2 + y^2)dy$, where c is the boundary	15	5	High	Very difficult	CO-4	Numerical

		in xy-plane of the area enclosed by the x-axis and the semi-circle $x^2 + y^2 = 1$in the upper half of xy-plane. **Soln.**  Here, $M = 2x^2 - y^2$, $N = x^2 + y^2$ $\frac{\partial M}{\partial y} = -2y, \frac{\partial N}{\partial x} = 2x$ $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 2x + 2y = 2(x + y)$ By Green's theorem, $\int_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\, dy$ $= \iint_R 2(x + y)\, dx\, dy$ $= \int_{y=0}^{1} \int_{x=0}^{\sqrt{1-y^2}} 2(x + y)\, dx\, dy$ $= 2 \int_{y=0}^{1} \left[\frac{x^2}{2} + xy\right]_0^{\sqrt{1-y^2}} dy$ $= \int_{y=0}^{1} \left[(1 - y^2) + 2y\sqrt{1 - y^2}\right] dy$						

		$= \left[y - \frac{y^3}{3} \right]_0^1 + 2 \int_{y=0}^{1} y\sqrt{1-y^2}\, dy$ Put $y = \sin\theta$, $dy = \cos\theta\, d\theta$. When $y = 0$, $\theta = 0$ and $y = 1$, $\theta = \pi/2$ $= 1 - \frac{1}{3} + 2 \int_0^{\pi/2} \sin\theta.\cos\theta.\cos\theta.\, d\theta$ $= \frac{2}{3} + 2 \int_0^{\pi/2} \sin\theta \cdot \cos^2\theta\, d\theta$ $= \frac{2}{3} + 2 \left[\frac{\cos^3\theta}{-3} \right]_0^{\pi/2}$ $= \frac{2}{3} + \frac{2}{3}$ $= \frac{4}{3}$ $\therefore \boxed{\int_C [Mdx + Ndy] = \frac{4}{3}}$ **Ans.**						
32	Unit 6	By using Green's theorem show that the area bounded by a simple closed curve c is given by $\frac{1}{2}\oint_c (xdy - ydx)$, hence	15	5	High	Very difficult	CO-4	Numerical

		deduced that the area of asteroid $x^{2/3} + y^{2/3} = a^{2/3}$. **Soln.** By Green's theorem in plane, we have $\int_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$ Here, $M = -y \Rightarrow \frac{\partial M}{\partial y} = -1$ and $N = x \Rightarrow \frac{\partial N}{\partial x} = 1$						

$\therefore$ Eqn. (1) becomes

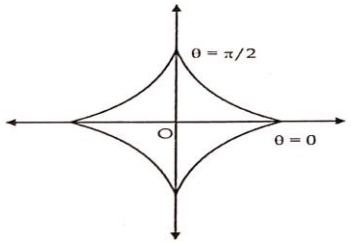
$$\int_C (-y\,dx + x\,dy) = \iint_R (1+1)\,dx\,dy = 2\iint_R dx\,dy = 2A$$

where A is the required area.

$$\therefore \quad \boxed{A = \frac{1}{2}\int_C (x\,dy - y\,dx)} \quad \textbf{Ans.}$$

Now, any point (x, y) on the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ is given by

$$x = a\cos^3\theta, \quad y = a\sin^3\theta$$



$$\Rightarrow dx = -3a\cos^2\theta\sin\theta\,d\theta, \quad dy = 3a\sin^2\theta\cos\theta\,d\theta$$

Hence, the area of the astroid

$$= \frac{1}{2}\cdot 4\int_0^{\pi/2} [a\cos^3\theta \cdot 3a\sin^2\theta\cos\theta - a\sin^3\theta \cdot 3a\cos^2\theta(-\sin\theta)]\,d\theta$$

$$= 6a^2\int_0^{\pi/2} (\cos^4\theta\sin^2\theta + \sin^4\theta\cos^2\theta)\,d\theta$$

$$= 6a^2\left(\frac{3.1.1}{6.4.2}\frac{\pi}{2} + \frac{3.1.1}{6.4.2}\frac{\pi}{2}\right)$$

[$\because$ By prop. of Beta function]

$$= 6a^2\frac{\pi}{16} = \frac{3}{8}\pi a^2$$

$\therefore$ $\boxed{\text{The area of the astroid} = \frac{3}{8}\pi a^2}$ **Ans.**